

Weekend of Code 9.0

Turing's Playground Abstract

Team: Slip off Tongue

Team Members:

Member 1: Durgesh Patel, Registration No. 2025ca035, CSED, GitHub:
<https://github.com/durgeshptl> (**Team Leader**)

Member 2: Anurag Gupta, Registration No.2025ca018, CSED, GitHub:
<https://github.com/Anurag2153>

Member 3: Mritunjay Kumar Singh , Registration No. 2025ca058, CSED, GitHub:
GitHub: <https://github.com/MritunjaySingh28>

Problem Statement ID: A01

Problem Statement:

To design and implement an intelligent system capable of automatically detecting and classifying toxic user-generated comments in online platforms.

Description:

This project focuses on building a machine learning-based system to detect toxic comments such as insults, threats, and obscene language using NLP and supervised learning techniques.

Outline of the Project:

- Data collection and preprocessing
- Data mining
- Data cleaning
- Data visualization
- TF-IDF vectorization
- Multi-label classification using Logistic Regression, Naive-Bayesian/BERT
- Classification report
- Web application development using Flask

Libraries Used:

- Python
- Flask
- Scikit-learn
- NLTK
- Pandas and more

Highlights:

- Multi-label toxic comment detection
- Percentage-based toxicity output
- User-friendly web interface